# A gentle introduction to the calculus of variations

(These slides: bit.ly/intro-calc-var)

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# Stuff we'll do here:

A vague outline of the talk:

- A motivating problem
- An easier problem
- A general approach,
- applied to the first problem

Math stuff we'll talk about:

- Functionals
- Differentiation under the integral sign
- Integration by parts
- Some differential equations
- Lagrange multipliers

# A motivating problem

# The catenary

## A motivating problem

What function describes the shape taken by a hanging chain?

# Functionals, variations, and “wiggles”

# Functionals

What is a function, anyway?

- Some kind of rule associating inputs with outputs
- $f : \mathbb{R} \to \mathbb{R}$

A functional is a function whose inputs are functions

- $J : C^2 \to \mathbb{R}$
- (or, y'know, whatever function space)
- Most functionals we care about end up being definite integrals:

$$J[y] = \int_a^b f(x, y, y')\, dx$$

(If I were naming this subject, I would call it "the calculus of functionals," but nobody asked me)

# What if I want to minimize a functional?

We want to take some kind of derivative

- (because derivatives are a nice way to handle extrema)

**Definition (kinda)**

The derivative is a device that measures
the effect on the output $dy$
of small changes in the input. $dx$

So how should we deal with this for functionals, where the inputs are themselves functions?

## Variations and "wiggles"

A variation is a function $\eta(x)$ who is:

- not too wild: piecewise $C^2$ or maybe even piecewise $C^1$
- zero at endpoints: $\eta(x_1) = \eta(x_2) = 0$

A "wiggle" is the result of adding some amount of variation to a base function:

- $w(x) = y(x) + a\,\eta(x)$
- "A one-parameter family of curves" that includes original $y(x)$ (when $a = 0$)

> "Wiggles" seem like a good way to capture the idea of "small changes to inputs" in the context of functionals.

# Derivatives of functionals

Instead of feeding the functional $J$ the original function $y$, let's feed it a wiggle $w = y + a\eta$:

$$J[w] = J[y + a\eta] = \int_{x_1}^{x_2} f(x, y + a\eta, y' + a\eta')\ dx$$

This is truly a function of $a$!

$$J_y(a) = \int_{x_1}^{x_2} f(x, y + a\eta, y' + a\eta')\ dx$$

I can just regular-calculus-differentiate w.r.t. $a$! (I hope)

## Our strategy

- Write down a functional: $J[y] = \int_{x_1}^{x_2} f(x, y, y')\ dx$
- Construct a "wiggle" by adding "variation:" $w(x) = y(x) + a \cdot \eta(x)$
- Apply the functional to the wiggle: $J(a) = J[y + a\eta] = \int_{x_1}^{x_2} f(x, y + a\eta, y' + a\eta')\ dx$
- Take some kind of derivative: $\frac{d}{da} J(a)$
- Assume $y$ (corresponding to $a = 0$) is a minimizer:
  - Assert $J'(0) = 0$
- Try to conclude something interesting about $y$

# "Some kind of derivative," eh?

Let's think a bit harder about how this, theoretically, just-plain-calculus derivative $\frac{d}{da}J(a)$ actually works.

# Differentiating under the integral sign

$$\frac{d}{da}J(a) = \frac{d}{da}\left[\int_{x_1}^{x_2} f(x, y + a\eta, y' + a\eta')\ dx\right]$$

Oh no, all of my $a$'s are inside this integral, whatever shallst I do

**Leibniz integral rule** (sometimes this is called Feynman's integral trick but he's kind of a jerk)

As long as the integrand is continuous,

$$\frac{d}{da}\left[\int_{x_1}^{x_2} g(x, a)\ dx\right] = \int_{x_1}^{x_2}\left[\frac{\partial}{\partial a}g(x, a)\right]\ dx$$

(Note: $d/da$ outside becomes $\partial/\partial a$ inside)

- This is basically Fubini's theorem and/or Clairault's theorem

# Okay, so,

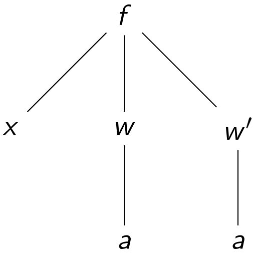
$$\frac{d}{da}J(a) = \frac{d}{da}\left[\int_{x_1}^{x_2} f(x, y + a\eta, y' + a\eta')\ dx\right]$$
$$= \int_{x_1}^{x_2} \left[\frac{\partial}{\partial a} f(x, y + a\eta, y' + a\eta')\right]\ dx$$

or in other words

$$= \int_{x_1}^{x_2} \left[\frac{\partial}{\partial a} f(x, w, w')\right]\ dx$$

I guess let's zoom in on this integrand $\dfrac{\partial f}{\partial a}$.

# Chain rule: How does $f$ depend on $a$?




$$\frac{\partial f}{\partial a} = \frac{\partial f}{\partial w}\frac{\partial w}{\partial a} + \frac{\partial f}{\partial w'}\frac{\partial w'}{\partial a}$$

- Well, $\dfrac{\partial w}{\partial a} = \eta$, and $\dfrac{\partial w'}{\partial a} = \eta'$
- so $\dfrac{\partial f}{\partial a} = \dfrac{\partial f}{\partial w} \cdot \eta + \dfrac{\partial f}{\partial w'} \cdot \eta'$
- and we're interested in $a = 0$ where $w = y + 0\eta$
- Thus,

$$\left.\frac{\partial f}{\partial a}\right|_{a=0} = \frac{\partial f}{\partial y} \cdot \eta + \frac{\partial f}{\partial y'} \cdot \eta'$$

# Zooming back out

$$
\begin{aligned}
\frac{d}{da}J(a) &= \frac{d}{da}\left[\int_{x_1}^{x_2} f(x, y + a\eta, y' + a\eta')\ dx\right] \\
&= \int_{x_1}^{x_2} \left[\frac{\partial}{\partial a} f(x, w, w')\right]\ dx \\
&= \int_{x_1}^{x_2} \left[\frac{\partial f}{\partial w} \cdot \eta + \frac{\partial f}{\partial w'} \cdot \eta'\right]\ dx \\
\left.\frac{d}{da}J(a)\right|_{a=0} &= \int_{x_1}^{x_2} \frac{\partial f}{\partial y} \cdot \eta + \frac{\partial f}{\partial y'} \cdot \eta'\ dx
\end{aligned}
$$

We're then asserting that this last integral is zero.

# Observations:

$$0 = \frac{d}{da} J(a)\bigg|_{a=0} = \int_{x_1}^{x_2} \frac{\partial f}{\partial y} \cdot \eta + \frac{\partial f}{\partial y'} \cdot \eta' \; dx$$

- These two summands look suspiciously similar.
- In the first, we have something multiplied against $\eta$;
- in the second, something multiplied against $\eta'$.
- Maybe we can exploit this structure.

# The Fundamental Lemma of the Calculus of Variations

# Lemma time

**The Fundamental Lemma of the Calculus of Variations (!!!)**

Suppose $m(x)$ is continuous. If $\int_{x_1}^{x_2} m(x) \cdot \eta(x)\ dx = 0$ for all variations $\eta$, then $m(x) \equiv 0$.

Observations:

- This is truly just the $L_2$ inner product $\langle m, \eta \rangle$
- So what this says is, the only thing that's orthogonal to every variation is the zero function
- (This restatement makes it more believable to me)

## Proof.

Suppose not: say WLOG that $m(x^*) > 0$. Since $m$ is continuous, there exists an interval $[c, d]$ around $x^*$ on which $m(x) > 0$.

Let $\eta(x) = \begin{cases} 0 & x \notin [c,d] \\ (x-c)(d-x) & x \in [c,d] \end{cases}$. $\eta$ is a variation.

$$\begin{aligned} 0 &= \int_{x_1}^{x_2} m(x) \cdot \eta(x)\ dx \\ &= \underbrace{\int_{x_1}^{c} m(x) \cdot \eta(x)\ dx}_{0} + \underbrace{\int_{c}^{d} m(x) \cdot \eta(x)\ dx}_{>0} + \underbrace{\int_{d}^{x_2} m(x) \cdot \eta(x)\ dx}_{0} \end{aligned}$$

So we have $0 > 0$, and we've reached our **contradiction.** □

# The Euler-Lagrange equation

# Reminder of our strategy

- Write down a functional: $J[y] = \int_{x_1}^{x_2} f(x, y, y')\ dx$
- Construct a "wiggle" by adding "variation:" $w(x) = y(x) + a \cdot \eta(x)$
- Apply the functional to the wiggle: $J(a) = J[y + a\eta] = \int_{x_1}^{x_2} f(x, y + a\eta, y' + a\eta')\ dx$
- Take some kind of derivative: $\frac{d}{da} J(a)$
- Assume $y$ (corresponding to $a = 0$) is a minimizer:
  - Assert $J'(0) = 0 \leftarrow$ We are here
- Try to conclude something interesting about $y$

# Ready for some integration by parts?

$$\int_{x_1}^{x_2} \frac{\partial f}{\partial a}\, dx = \int_{x_1}^{x_2} \frac{\partial f}{\partial y} \cdot \eta + \frac{\partial f}{\partial y'} \cdot \eta' \; dx$$

$$u = \tfrac{\partial f}{\partial y'} \qquad v = \eta$$

$$du = \tfrac{d}{dx}\left(\tfrac{\partial f}{\partial y'}\right) dx \quad dv = \eta' \; dx$$

$$\left[\tfrac{\partial f}{\partial y'} \cdot \eta\right|_{x_1}^{x_2} - \int_{x_1}^{x_2} \eta \cdot \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) dx$$

$$= \int_{x_1}^{x_2} \frac{\partial f}{\partial y} \cdot \eta - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) \cdot \eta \; dx$$

$$= \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right)\right) \cdot \eta \; dx = 0 \qquad \text{FLCOV!!!}$$

# Conclude something interesting about $y$

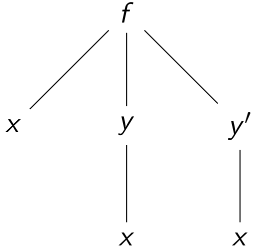
## Euler-Lagrange equation

If $y$ is a minimizer for the functional $J[y] = \int_{x_1}^{x_2} f(x, y, y')\, dx$, then $y$ satisfies the equation

$$\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = 0. \quad \left(\text{Or, } \frac{\partial f}{\partial y} = \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right).\right)$$

# Toward a special case

This $\frac{d}{dx}$ business is weird. Let's think about $\frac{df}{dx}$.




$$\begin{aligned}\frac{df}{dx} &= \frac{\partial f}{\partial y}\cdot\frac{dy}{dx}+\frac{\partial f}{\partial y'}\cdot\frac{dy'}{dx}+\frac{\partial f}{\partial x}\\ &= \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right)\cdot y'+\frac{\partial f}{\partial y'}\cdot\frac{d}{dx}(y')+\frac{\partial f}{\partial x}\end{aligned}$$

Apply the product rule backward!

$$= \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\cdot y'\right)+\frac{\partial f}{\partial x}$$

$$\frac{df}{dx}-\frac{d}{dx}\left(\frac{\partial f}{\partial y'}\cdot y'\right)=\frac{\partial f}{\partial x}$$

# Toward a special case

So, suppose that $\frac{\partial f}{\partial x} = 0$.

- In other words, the integrand of the functional doesn't depend explicitly on $x$
- (This happens frequently because many situations we'd care about are autonomous slash time-invariant)

Then:

$$\frac{df}{dx} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'} \cdot y'\right) = \frac{\partial f}{\partial x} = 0$$

$$\frac{d}{dx}\left(f - \frac{\partial f}{\partial y'} \cdot y'\right) = 0$$

$$f - \frac{\partial f}{\partial y'} \cdot y' = C$$

# Conclude something interesting about $y$, part deux

## Euler-Lagrange equation

If $y$ is a minimizer for the functional $J[y] = \int_{x_1}^{x_2} f(x, y, y')\, dx$, then $y$ satisfies the equation

$$\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = 0. \quad \left(\text{Or, } \frac{\partial f}{\partial y} = \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right).\right)$$

## Special case: Beltrami identity

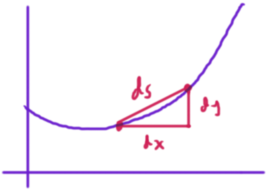
If $\frac{\partial f}{\partial x} = 0$, then $y$ satisfies the equation

$$f - \frac{\partial f}{\partial y'} \cdot y' = C.$$

# Some examples

# Example: arc length

(Fun toy: Wolfram Alpha "arc length")
Put in a function, get out its length (between two $x$-values)
Let's do a little calculus: slice-approximate-integrate




$$
\begin{aligned}
ds^2 &= dx^2 + dy^2 \\
&= \left(1 + \frac{dy^2}{dx^2}\right) dx^2 \\
ds &= \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \\
&= \sqrt{1 + (y')^2}\, dx
\end{aligned}
$$

# Example: Arc length

Cool, so $J[y] = \int ds = \int_{x_1}^{x_2} \sqrt{1+(y')^2}\, dx = \int_{x_1}^{x_2} f(x,y,y')\, dx$

Some notes:

- Most functionals we care about end up being definite integrals
- Many functionals we care about don't depend explicitly on $x$

## An easier problem

What function connecting two given points $(x_1, y_1)$ and $(x_2, y_2)$ has the minimal arc length? Note: You can at this point just apply the Beltrami identity, but let's have some fun.

# The arc length of a wiggle

Explore on Desmos!

Upshot:

- $J[w] = J[y + a\eta] = \int_{x_1}^{x_2} \sqrt{1 + (y' + a\eta')^2}\, dx$
- This is truly a function of $a$!

$$J(a) = \int_{x_1}^{x_2} \sqrt{1 + (y' + a\eta')^2}\, dx$$

- So I can take a derivative w.r.t. $a$!
- If $y$ (corresp. $a = 0$) is a minimizer, then $J'(0) = 0$

# Calculating toward $J'(0) = 0$

$$\begin{aligned}
\frac{d}{da} J(a) &= \int_{x_1}^{x_2} \frac{\partial}{\partial a} \left[1 + (y' + a\eta')^2\right]^{1/2} \, dx \\
&= \int_{x_1}^{x_2} \frac{1}{2} \left[1 + (y' + a\eta')^2\right]^{-1/2} \cdot 2(y' + a\eta') \cdot \eta' \, dx \\
&= \int_{x_1}^{x_2} \frac{y' + a\eta'}{[1 + (y' + a\eta')^2]^{1/2}} \cdot \eta' \, dx
\end{aligned}$$

When $a = 0$,

$$0 = \left[\frac{d}{da} J(a) \right|_{a=0} = \int_{x_1}^{x_2} \frac{y'}{[1 + (y')^2]^{1/2}} \cdot \eta' \, dx$$

# All good lemmas have corollaries

## The Fundamental Lemma of the Calculus of Variations (!!!)

Suppose $m(x)$ is continuous. If $\int_{x_1}^{x_2} m(x) \cdot \eta(x)\ dx = 0$ for all variations $\eta$, then $m(x) \equiv 0$.

## Corollary

Suppose $M(x)$ is continuous. If $\int_{x_1}^{x_2} M(x) \cdot \eta'(x)\ dx = 0$ for all variations $\eta$, then $M(x)$ is a constant function.

## Proof.

Integrate by parts! $\square$

# Consequences of $J'(0) = 0$

$$\int_{x_1}^{x_2} \frac{y'}{[1+(y')^2]^{1/2}} \cdot \eta' \, dx = 0$$

Apply the (corollary of the) Fundamental Lemma:

$$\frac{y'}{[1+(y')^2]^{1/2}} = C$$

Algebra time!

$$\begin{aligned} y' &= C[1+(y')^2]^{1/2} \\ (y')^2 &= C^2[1+(y')^2] \\ (y')^2 &= C^2 + C^2(y')^2 \\ (1-C^2)(y')^2 &= C^2 \\ (y')^2 &= \frac{C^2}{1-C^2}, \text{ so } y' = K \end{aligned}$$

# Conclusion

The shortest path between two points is a straight line!

# Conclusion

~~The shortest path between two points is a straight line!~~
If there is a shortest path between two points, it has to be a straight line!


(sorry / not sorry)

BTTF logo generator

# More precisely formulating the problem

- Often formulating the problem requires some physics
- Things generally want to minimize energy
- A chain is suspended from two points
- and it has a certain length

**The catenary problem**

What function passing through two specified points $(x_1, y_1)$ and $(x_2, y_2)$, with a specified length $\ell$, minimizes energy?

# We need a functional

- Often writing down the functional requires some physics
- "Energy" in this case is probably gravitational potential energy
- Slice-approximate-integrate: A little bit of chain is kinda like a point mass
- GPE $= mgh = (\rho\ ds)gh = (\rho g)h\ ds = (\rho g)y\sqrt{1+(y')^2}\ dx$
- Since we're minimizing, we can forget about those constants:

$$J[y] = \int_{x_1}^{x_2} y\sqrt{1+(y')^2}\ dx$$

# Now we can use the Beltrami identity $f - \frac{\partial f}{\partial y'} \cdot y' = C$

$$f(y, y') = y\sqrt{1 + (y')^2}; \quad \frac{\partial f}{\partial y'} = y \cdot \tfrac{1}{2}(1 + (y')^2)^{-1/2} \cdot 2y'$$

$$= \frac{y\ y'}{\sqrt{1 + (y')^2}}$$

$$y\sqrt{1 + (y')^2} - \frac{y(y')^2}{\sqrt{1 + (y')^2}} = C$$

$$y(1 + (y')^2) - y(y')^2 = C\sqrt{1 + (y')^2}$$

$$y + y(y')^2 - y(y')^2 = C\sqrt{1 + (y')^2}$$

$$y^2 = C^2\ (1 + (y')^2)$$

# $y^2 = C^2\,(1 + (y')^2)$

This is a separable DE

$$\begin{aligned} y^2 &= C^2\,(1 + (y')^2) \\ &= C^2 + C^2(y')^2 \\ y^2 - C^2 &= C^2(y')^2 \\ \frac{y^2}{C^2} - 1 &= (y')^2 \\ \left[\left(\frac{y}{C}\right)^2 - 1\right]^{1/2} &= y' \\ dx &= \frac{1}{\sqrt{(y/C)^2 - 1}}\,dy \end{aligned}$$

$\leftarrow$ You can integrate this but it's not nice

## $y^2 = C^2\left(1 + (y')^2\right)$

Differential equations is the study of dirty tricks
In a similar situation, the trick is a trigonometric substitution toward a parametric solution – What if $y' = \tan\theta$?

$$
\begin{aligned}
y^2 &= C^2(1 + tan^2\theta) \\
&= C^2 \sec^2\theta \\
y &= C\sec\theta \\
\frac{dx}{d\theta} &= \frac{dx}{dy}\frac{dy}{d\theta} \\
&= \frac{1}{\tan\theta}\cdot C\sec\theta\tan\theta
\end{aligned}
$$

$$
\begin{aligned}
\frac{dx}{d\theta} &= C\sec\theta \\
x &= C\ln(\tan\theta + \sec\theta) \\
e^{x/C} &= \tan\theta + \sec\theta \\
e^{x/C} &= \tan\theta + y/C \\
e^{x/C} &= y' + y/C\ldots
\end{aligned}
$$

I think this works, but it feels like the wrong trick.

# A better trick: sinh and cosh!

$$\sinh x = \frac{e^x - e^{-x}}{2}; \qquad \cosh x = \frac{e^x + e^{-x}}{2}$$

Useful properties:

- $\frac{d}{dx} \sinh x = \cosh x$
- $\frac{d}{dx} \cosh x = \sinh x$
- $\cosh^2 x = 1 + \sinh^2 x$

## $y^2 = C^2\,(1 + (y')^2)$

So I think maybe $y = \cosh \frac{x}{C}$? Then $y' = \frac{1}{C} \sinh \frac{x}{C}$, and so

$$\begin{aligned}(1 + (y')^2) &= 1 + \frac{1}{C^2} \sinh^2 \frac{x}{C} \\ C^2\,(1 + (y')^2) &= C^2 + \sinh^2 \frac{x}{C}\end{aligned}$$

Wait, dang. Revised guess: $y = C \cosh \frac{x}{C}$. Then $y' = 1 \sinh \frac{x}{C}$, so

$$\begin{aligned}(1 + (y')^2) &= 1 + 1 \sinh^2 \frac{x}{C} \\ C^2\,(1 + (y')^2) &= C^2 + C^2 \sinh^2 \frac{x}{C} = C^2 \cosh^2 \frac{x}{C} \\ &= y^2 \quad \text{Yay!}\end{aligned}$$

## $y^2 = C^2\left(1 + (y')^2\right)$ has a solution $y = C\cosh(x/C)$

I feel like we need another constant, though:

- There's two endpoints $(x_1, y_1)$ and $(x_2, y_2)$, and I think each one should determine one parameter
- Also, our dirty trick didn't provide for a constant of integration

Note that our differential equation was autonomous (doesn't depend on $x$), so horizontal shifts don't affect the DE

$$y = C\cosh\left(\frac{x-h}{C}\right)$$

(Another fun dirty trick!)

# What about the length of the chain?

- Constrained optimization: Lagrange multipliers!
- In multivariable calculus, $\nabla f - \lambda \nabla g = 0$
- This generalizes: think of $\nabla$ as "whatever kind of derivative makes sense in context"
- Our constraint: arc length! $L[y] = \int \sqrt{1+(y')^2}\, dx = \ell$

$$\frac{d}{da}\int y\sqrt{1+(y')^2}\, dx - \lambda \cdot \frac{d}{da}\int \sqrt{1+(y')^2}\, dx = 0$$

$$\frac{d}{da}\int (y-\lambda)\sqrt{1+(y')^2}\, dx = 0$$

Apply Beltrami to our new integrand!

# Apply Beltrami to $f(y, y') = (y - \lambda)\sqrt{1 + (y')^2}$

That $\lambda$ isn't gonna do much in derivatives

$$(y - \lambda)^2 = C^2 \left(1 + (y')^2\right)$$

You can spot that $y - \lambda$ is taking the place of $y$, so

$$y = \lambda + C \cosh\left(\frac{x - h}{C}\right) \quad (!!)$$

This has the correct number of parameters ($\lambda,\ C,\ h$) to correspond to three free choices (two fixed endpoints plus a length)
(The values probably have to be solved numerically)
Explore on Desmos!

# Further resources

- A lovely old textbook by Bliss (1925)
    - full pdf; digitized version (decidedly incomplete!)
    - "Remarks," "activities," etc. are from me and my students
- A nice treatment of the catenary and some related problems
- Watch this space!